CH 11- Solutions & Colloids

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- 11.1 The Dissolution Process
- 11.2 Electrolytes
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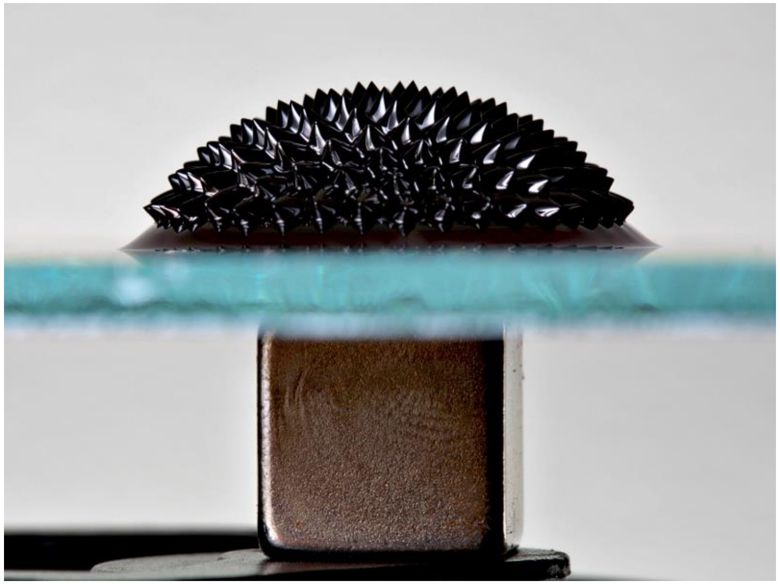
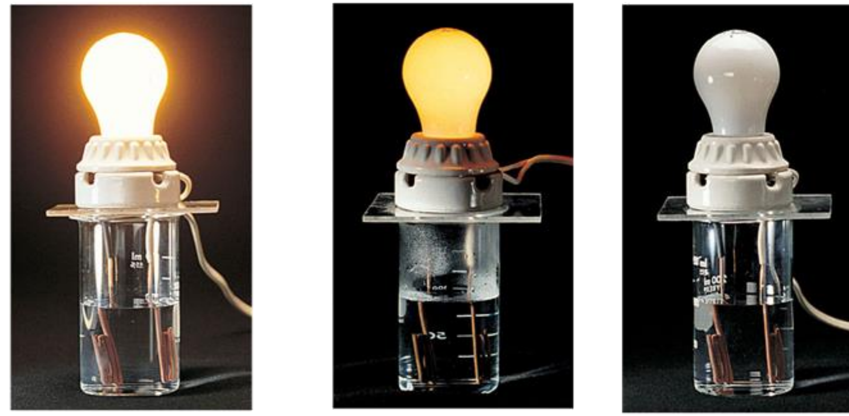
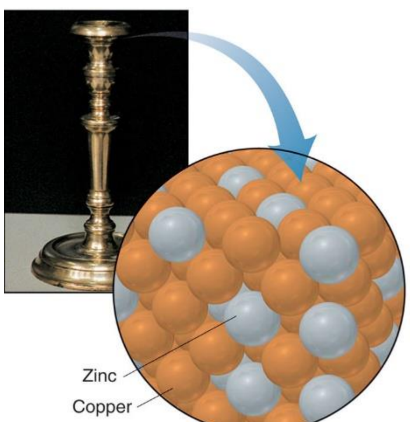
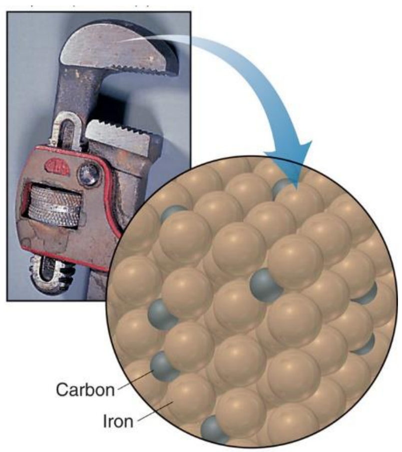




strong electrolyte weak electrolyte nonelectrolyte





A solution = A homogeneous mixture of two or more substances.

Solutions exhibit these defining traits:

- They are homogeneous; after a solution is mixed, it has the same composition at all points throughout (its composition is uniform).
- The physical state of a solution—solid, liquid, or gas—is typically the same as that of the solvent, as demonstrated by the examples in Table 11.1.
- The components of a solution are dispersed on a molecular scale; they consist of a mixture of separated solute particles (molecules, atoms, and/or ions) each closely surrounded by solvent species.
- The dissolved solute in a solution will not settle out or separate from the solvent.
- The composition of a solution, or the concentrations of its components, can be varied continuously (within limits determined by the *solubility* of the components, discussed in detail later in this chapter).

Different Types of Solutions

Solution	Solute	Solvent
air	$O_2(g)$	$N_2(g)$
soft drinks[1]	$CO_2(g)$	$H_2O(l)$
hydrogen in palladium	$H_2(g)$	$Pd(s)$
rubbing alcohol	$H_2O(l)$	$C_3H_8O(l)$ (2-propanol)
saltwater	$NaCl(s)$	$H_2O(l)$
brass	$Zn(s)$	$Cu(s)$

Table 11.1

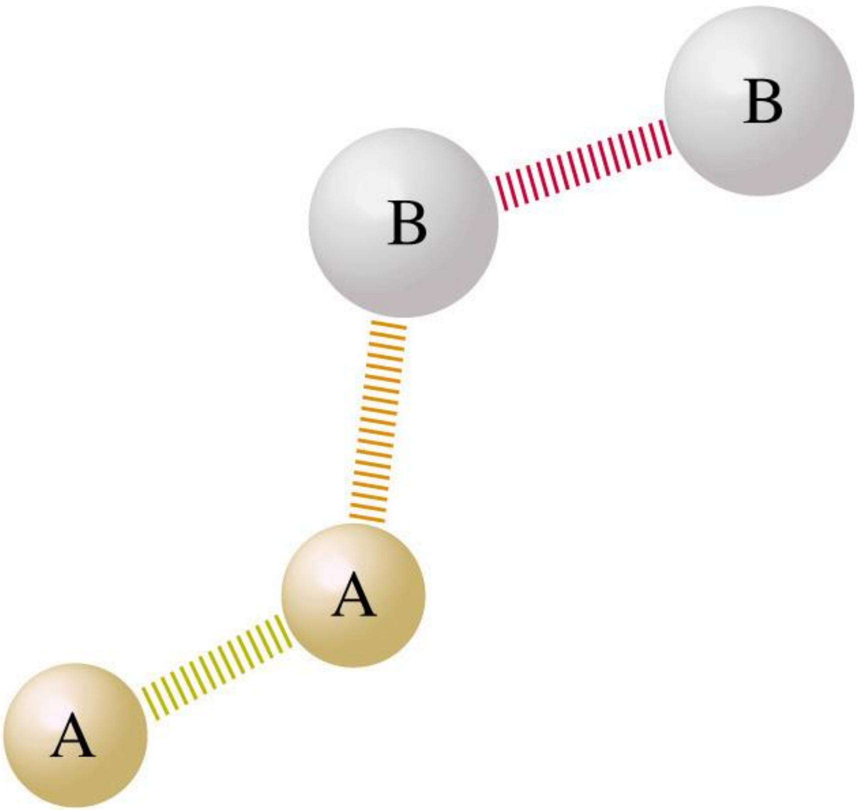
Assignment of Components as Solvent & Solute is somewhat arbitrary. Solute is usually lower in conc. Then solvent.

Solution Concentration

- Different measures of how much solute is dissolved in how much solvent.
  - small amount of solute: dilute solution
  - large amount of solute: concentrated solution
- The *solubility* (*S*) of a solute is the maximum amount that dissolves in a fixed quantity of solvent at a given temperature

NaCl solubility in Water = 39.12 g/100 mL (at 100 ^oC)
AgCl solubility in Water = 0.0021 g/100 mL (at 100 ^oC)

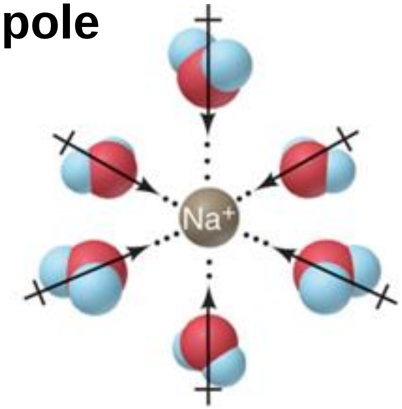
Intermolecular Forces and the Solution Process



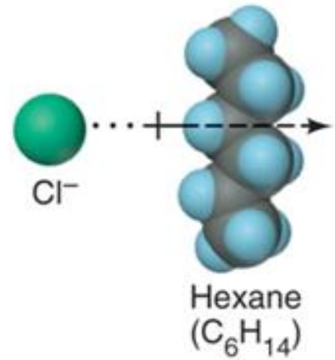
- Pure A has only A-A interactions.
- Pure B has only B-B interactions.
- Solution of A and B has A-A, B-B and A-B interactions.

Types of intermolecular forces in solutions

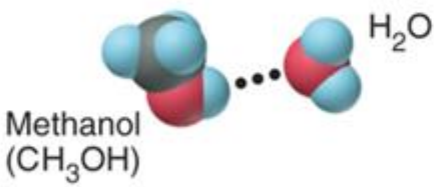
Ion-dipole



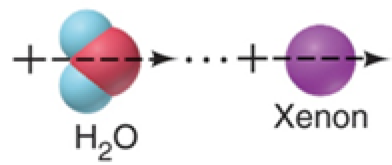
Ion-induced dipole



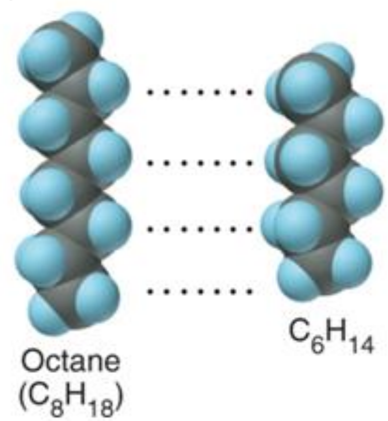
H bond



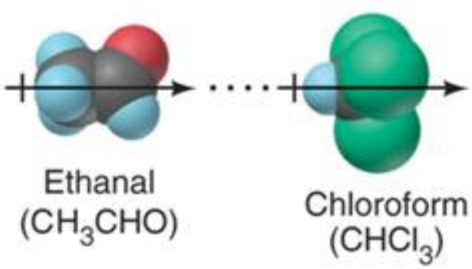
Dipole-induced dipole



Dispersion



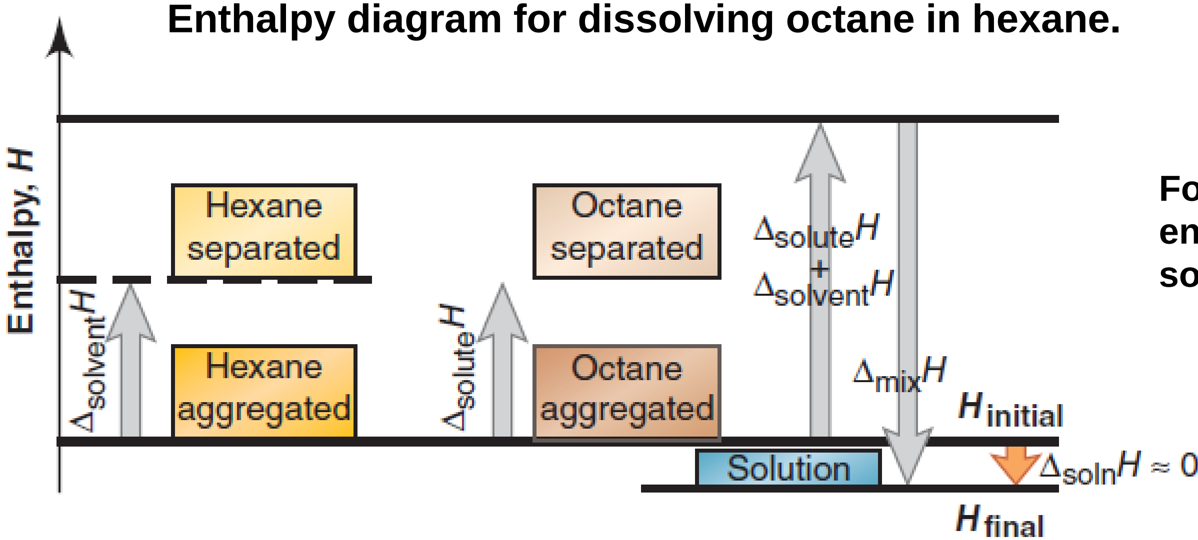
Dipole-dipole



Thermodynamics of Solution Formation

- An observed formation of a solution is an example of a **spontaneous process**, (dG is negative).
- Often there is an Enthalpy change, but not always, occurs as heat is absorbed or relased.
- dS can is positive in cases when the molecules are similar to each other (ethanol & water for example), but it can be negative in other cases when solvent molecules have to arrange themselves in particular ways to accommodate solutes.

Enthalpy diagram for dissolving octane in hexane.

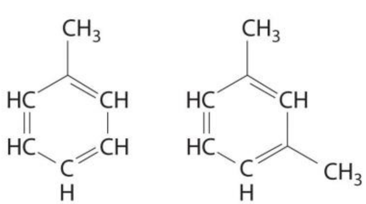


For octane, $\Delta_{soln}H$ is very small, but the entropy increase due to mixing is large, so octane dissolves (ΔG is negative)

$$\Delta H_{solution} \approx 0$$

ΔH is close to zero when forces between like and unlike molecules are very similar.

These mixtures are called ideal solutions.



Toluene m-Xylene

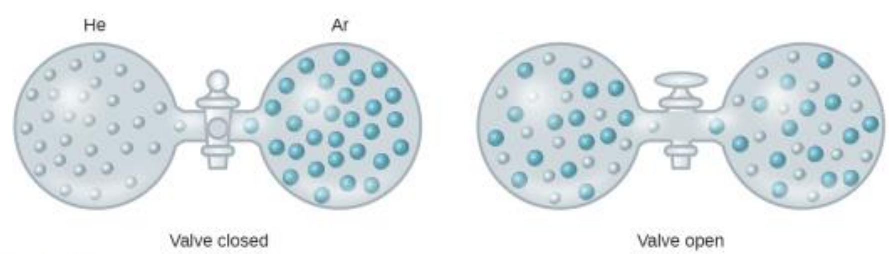


Figure 11.3 Samples of helium and argon spontaneously mix to give a solution.

- Three types of intermolecular attractive forces are relevant to the dissolution process: solute-solute, solvent-solvent, and solute-solvent.
- The formation of a solution may be considered by a hypothetical process in which energy is consumed to overcome solute-solute and solvent-solvent attractions (endothermic processes) and released when solute-solvent attractions are established (an exothermic process referred to as **solvation**).
- The relative magnitudes of the enthalpy changes associated with the hypothetical steps determine the process overall will release or absorb heat.
    - In some cases, solutions do not form because the process is too endothermic-entropy gain from solution formation does not compensate enough.

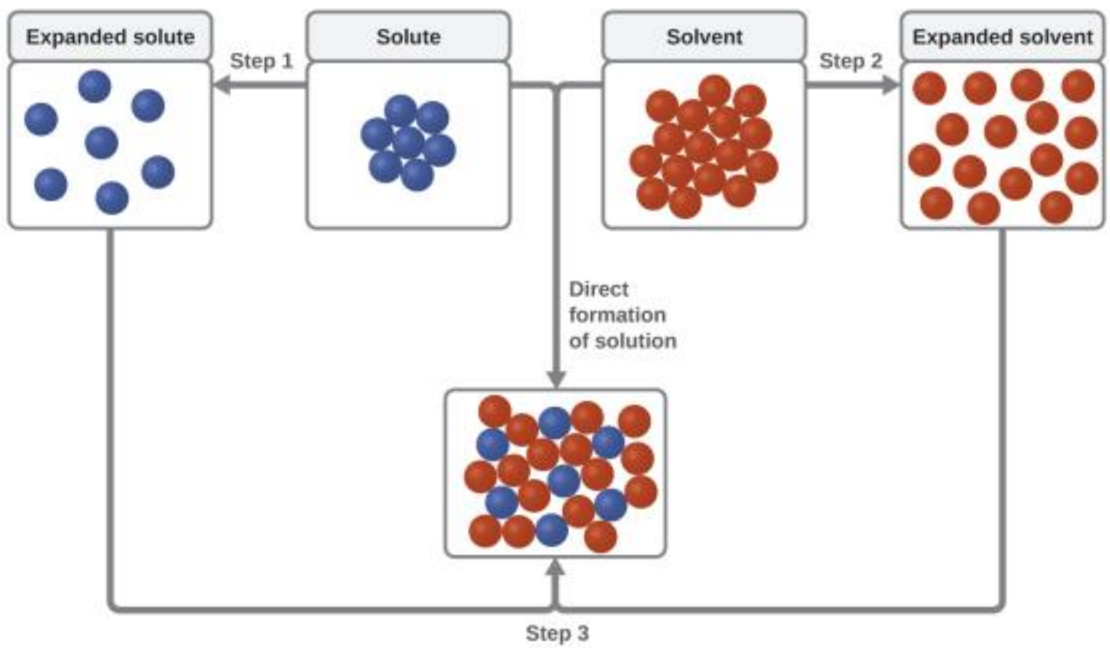
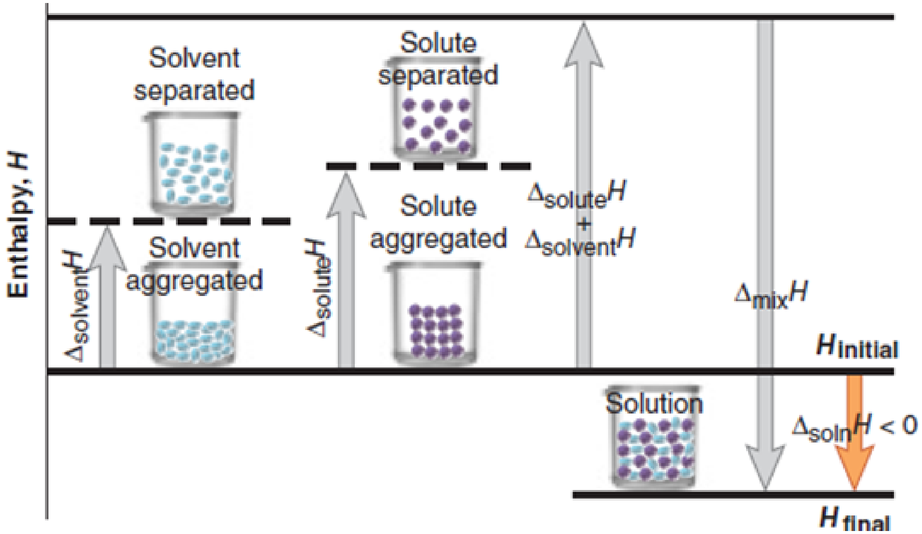


Figure 11.4 This schematic representation of dissolution shows a stepwise process involving the endothermic separation of solute and solvent species (Steps 1 and 2) and exothermic solvation (Step 3).



A. Exothermic solution process

$\Delta H_{mix} > \Delta H_{solute} + \Delta H_{solvent}$

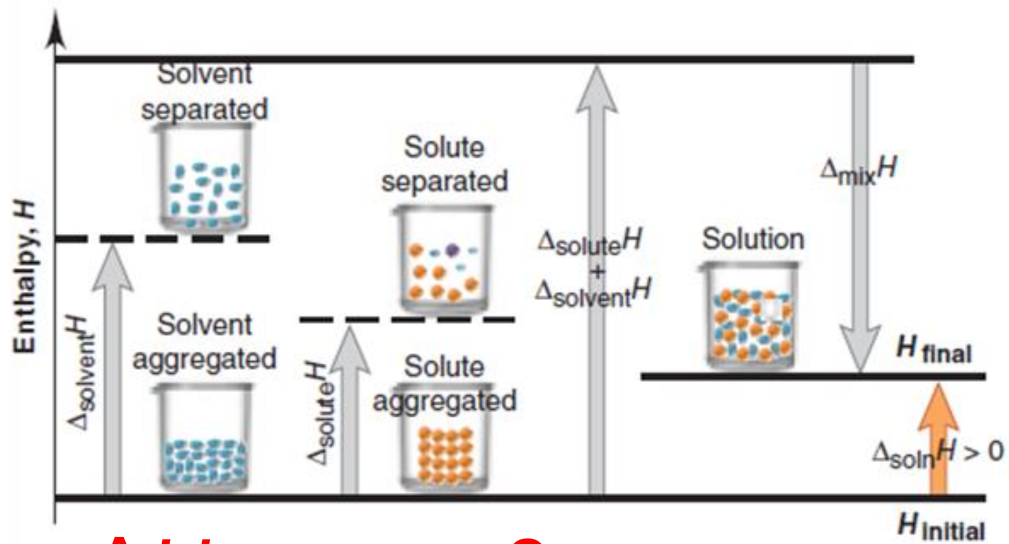
$\Delta H_{soln} < 0$

Forces between unlike molecules are stronger than are those between like molecules.

Overall exothermic $\Delta H_{solution} < 0$, and $\Delta S > 0$

Enthalpy components of the heat of solution.

B. Endothermic solution process
$\Delta H_{mix} < \Delta H_{solute} + \Delta H_{solvent}$
$\Delta H_{soln} > 0$



Overall endothermic: $\Delta H_{solution} > 0$

Forces between unlike molecules are weaker than are those between like molecules.

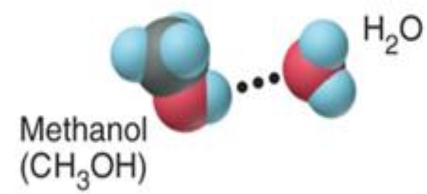
Dissolution of ammonium nitrate, NH_4NO_3, in water is endothermic.

Types of Solutions: Intermolecular Forces and Solubility

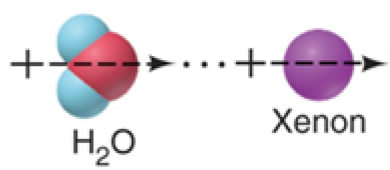
Dual Polarity and Effects on Solubility

- **Alcohols are organic compounds that have *dual polarity*.**
  - **The general formula for an alcohol is $CH_3(CH_2)_nOH$.**
- **The –OH group of an alcohol is *polar*.**
  - **It interacts with water through *H bonds* and**
  - **with hexane by through weak *dipole-induced dipole forces*.**
- **The hydrocarbon portion is *nonpolar*.**
  - **It interacts through weak *dipole-induced dipole forces* with water**
  - **and by *dispersion forces* with hexane.**

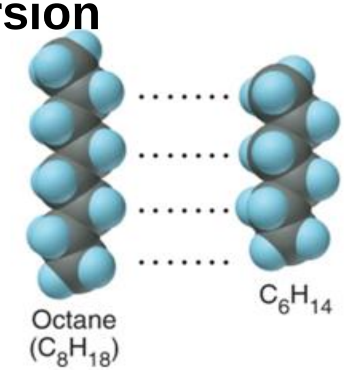
H bond



Dipole-induced dipole



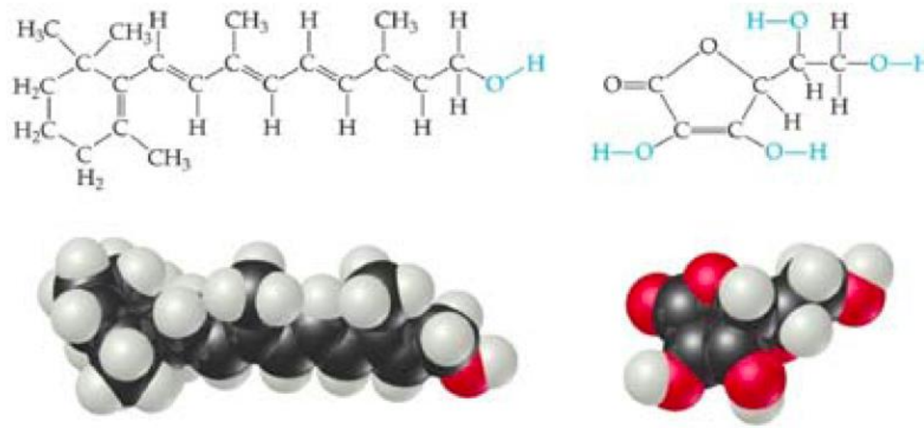
Dispersion



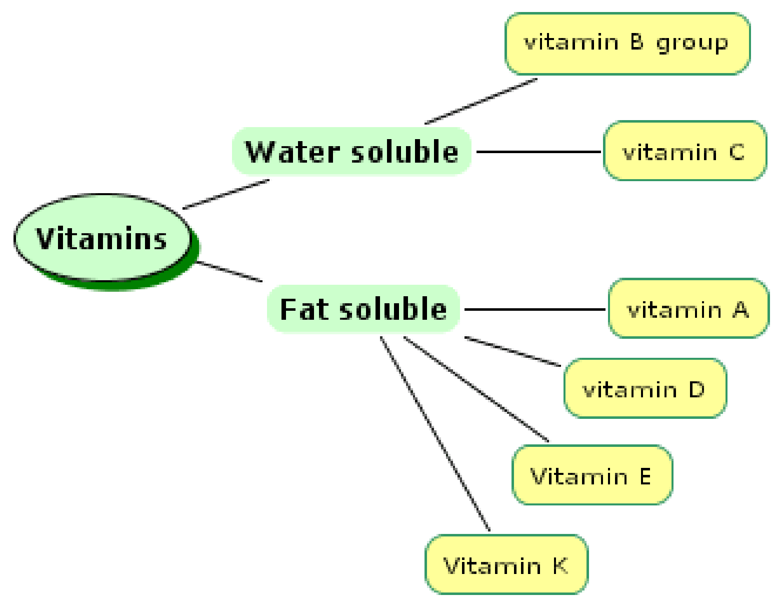
“Like dissolves like”

Polar compounds dissolve in polar solvents. → Water

Non-polar compounds dissolve in non-polar solvents. → Hexanes

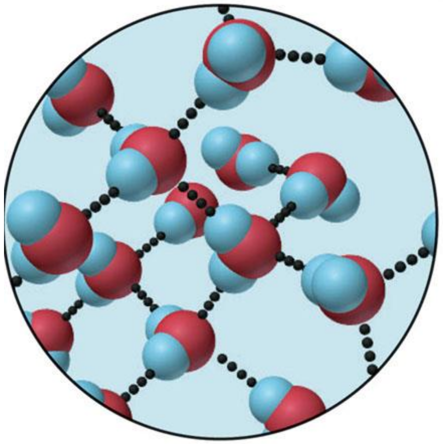


Vitamin A Vitamin C

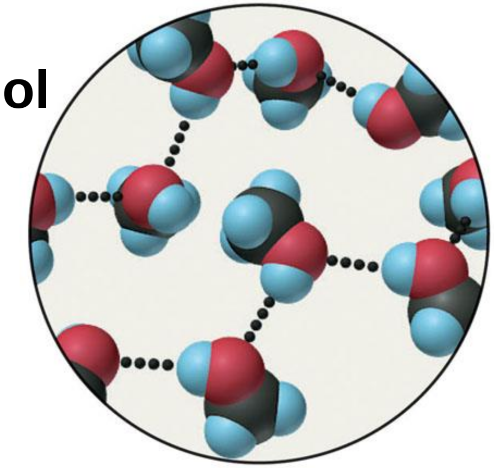


Like dissolves like: Liquid Solutions

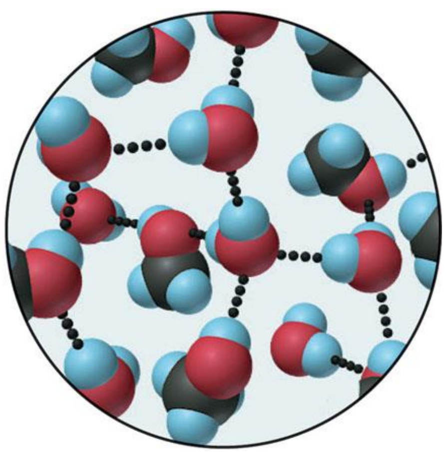
H_2O



Methanol
CH_3OH



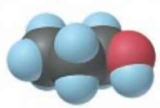
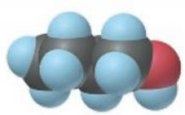
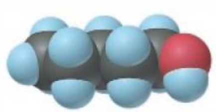
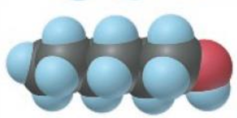
Both H_2O and methanol have H bonds between their molecules.



A solution of water and methanol.

- Some liquids may be mixed in any proportions to yield solutions; in other words, they have infinite mutual solubility and are said to be **miscible**.
    - Miscible liquids are typically those with very similar polarities.

TABLE 12.1 Solubility* of a Series of Alcohols in Water and in Hexane

Alcohol	Model	Solubility in Water	Solubility in Hexane
CH_3OH (methanol)		∞	1.2
CH_3CH_2OH (ethanol)		∞	∞
$CH_3(CH_2)_2OH$ (propan-1-ol)		∞	∞
$CH_3(CH_2)_3OH$ (butan-1-ol)		1.1	∞
$CH_3(CH_2)_4OH$ (pentan-1-ol)		0.30	∞
$CH_3(CH_2)_5OH$ (hexan-1-ol)		0.058	∞

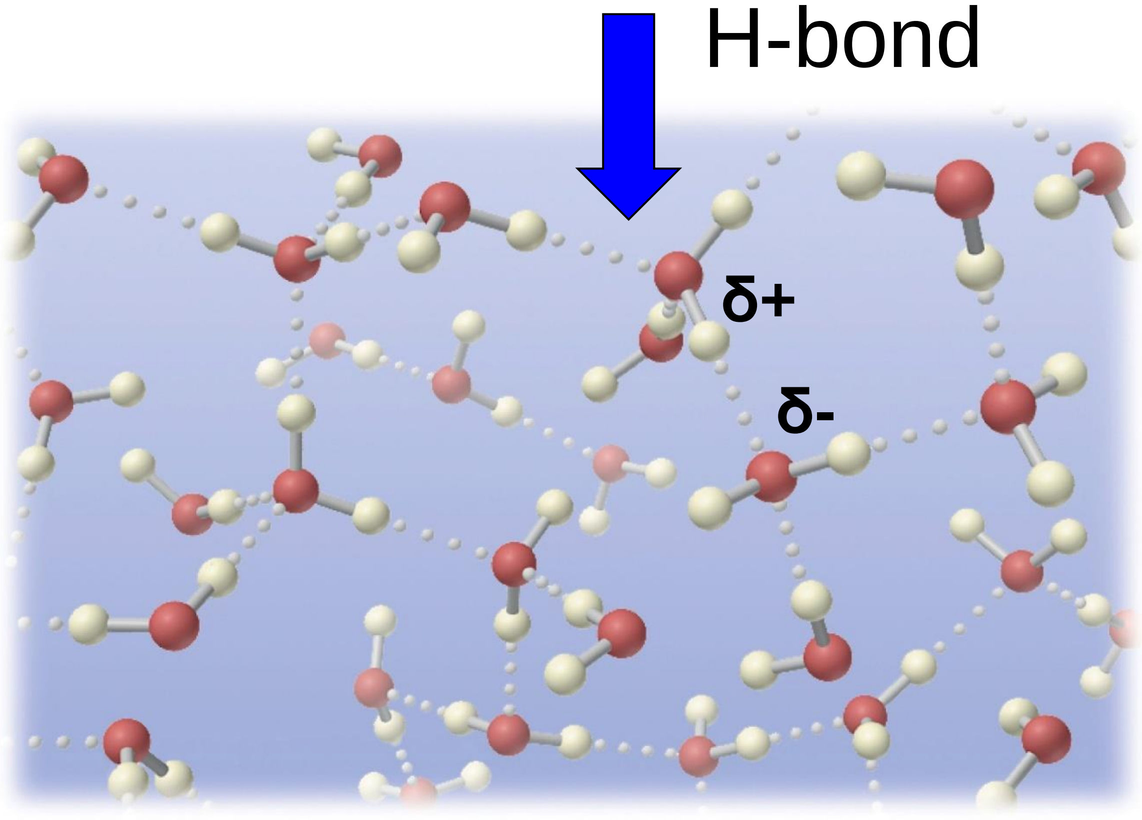
*Expressed in mol alcohol/1000 g solvent at 20°C.

- Two liquids that do not mix to a appreciable extent are called **immiscible**.
- Separate layers are formed when immiscible liquids are poured into the same container

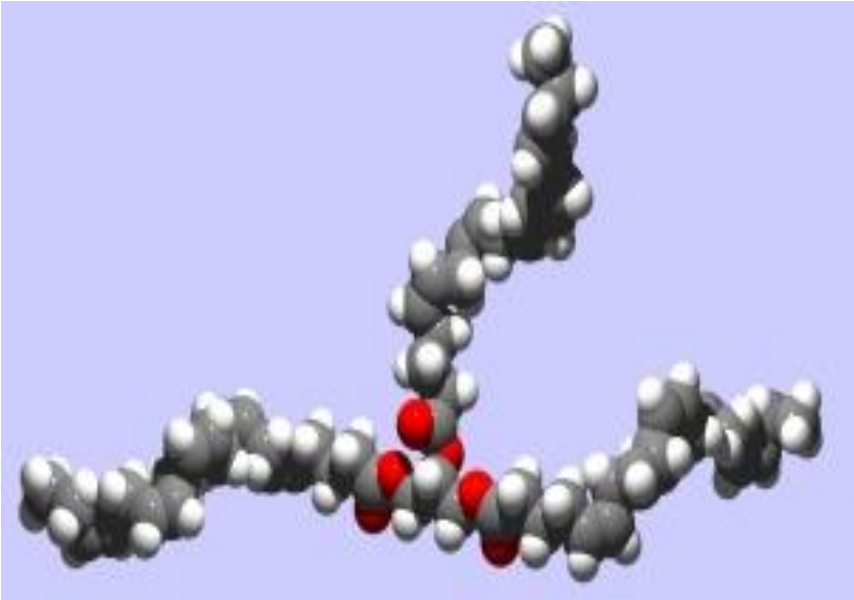
Heterogeneous Liquid Mixtures

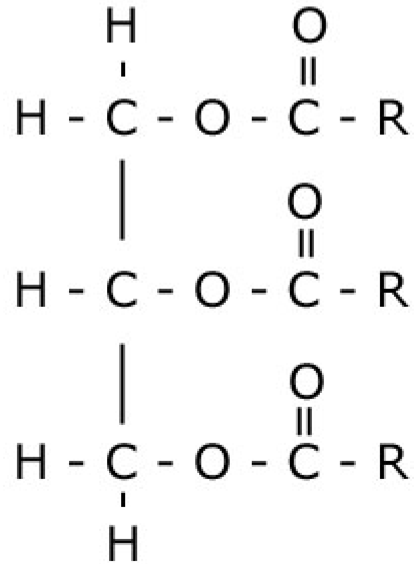
Small amts of oil in water phase could be detected, and small amounts of water would be found in oil phase.

Figure 11.14 Water and oil are immiscible. Mixtures of these two substances will form two separate layers with the less dense oil floating on top of the water. (credit: “Yortw”/Flickr)



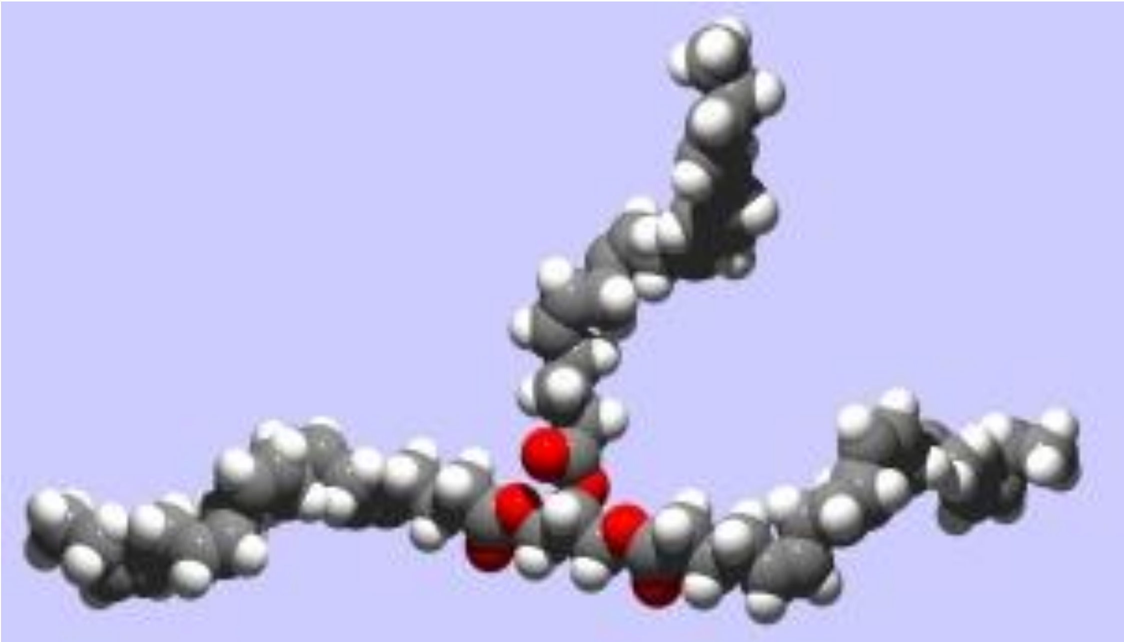
- Water forms a net work of strong intermolecular hydrogen bonds.

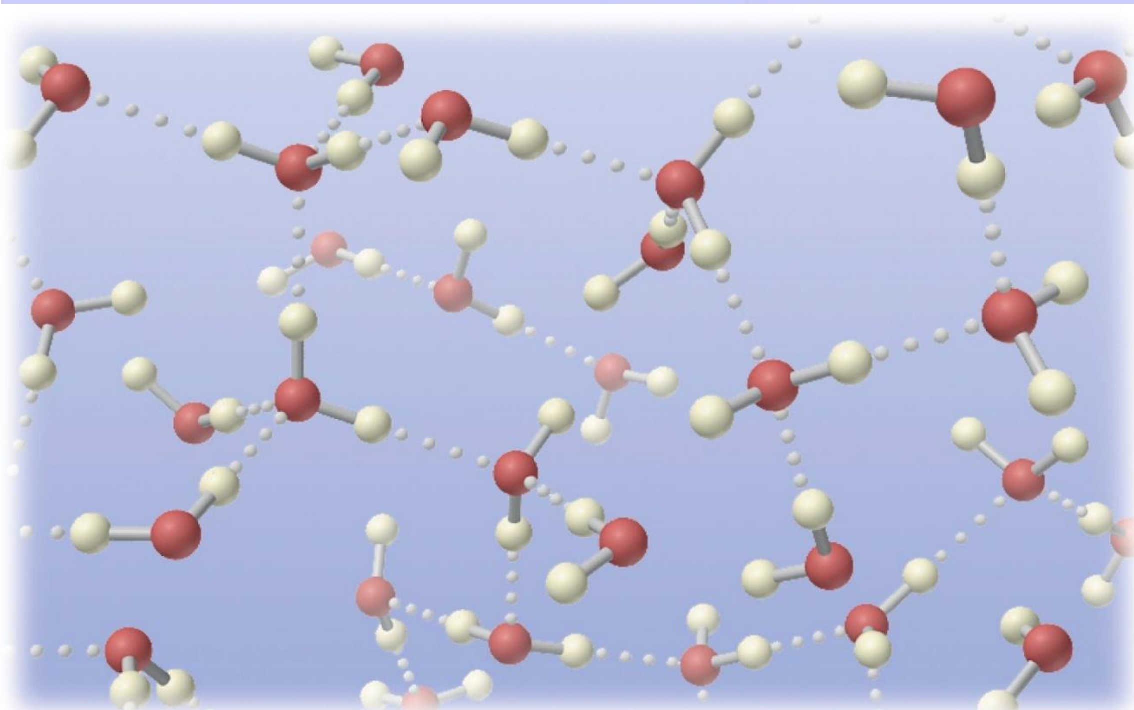




$R = CH_3(CH_2)_n$ (n = 14-22)

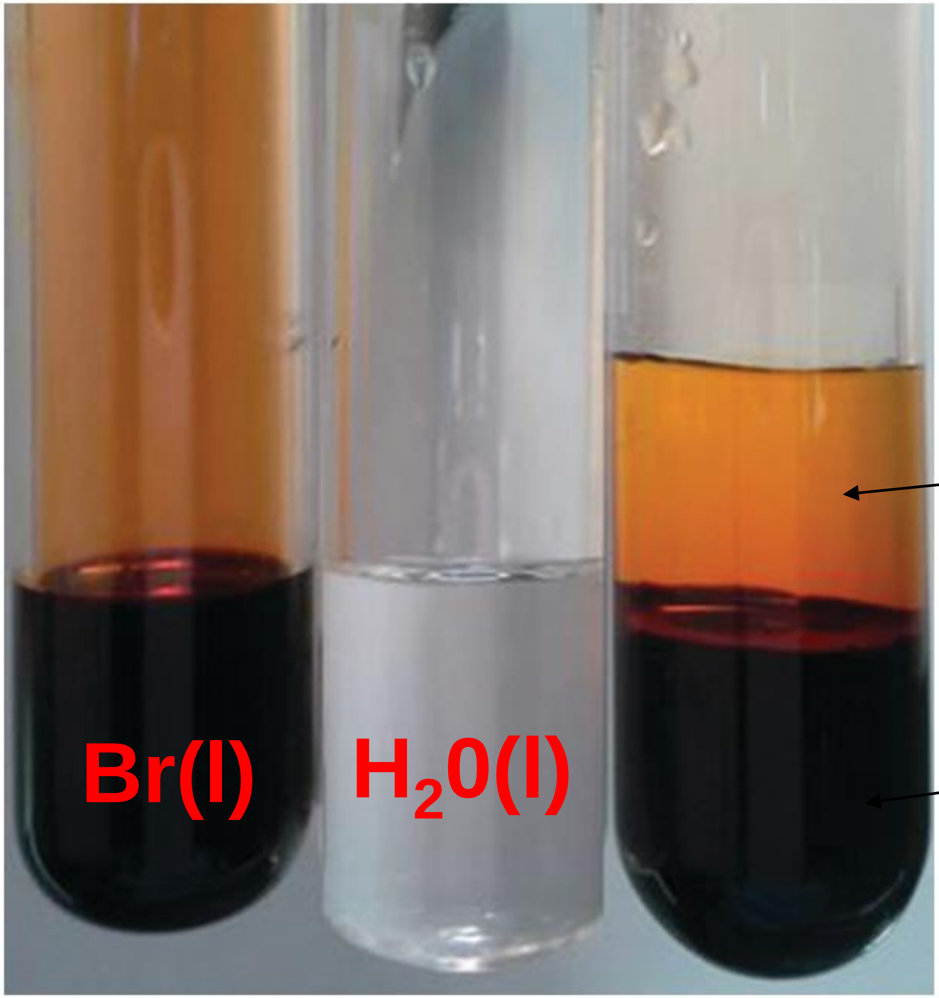
- Vegetable oil is composed of triglycerides (esters).
- Long alkyl chains cannot form hydrogen bonds





- Water/oil interactions are much weaker than water/water interactions.
- Heterogeneous mixture.

Two liquids, such as bromine and water, that are of *moderate* mutual solubility are said to be **partially miscible**.



Some Bromine in lots of Water

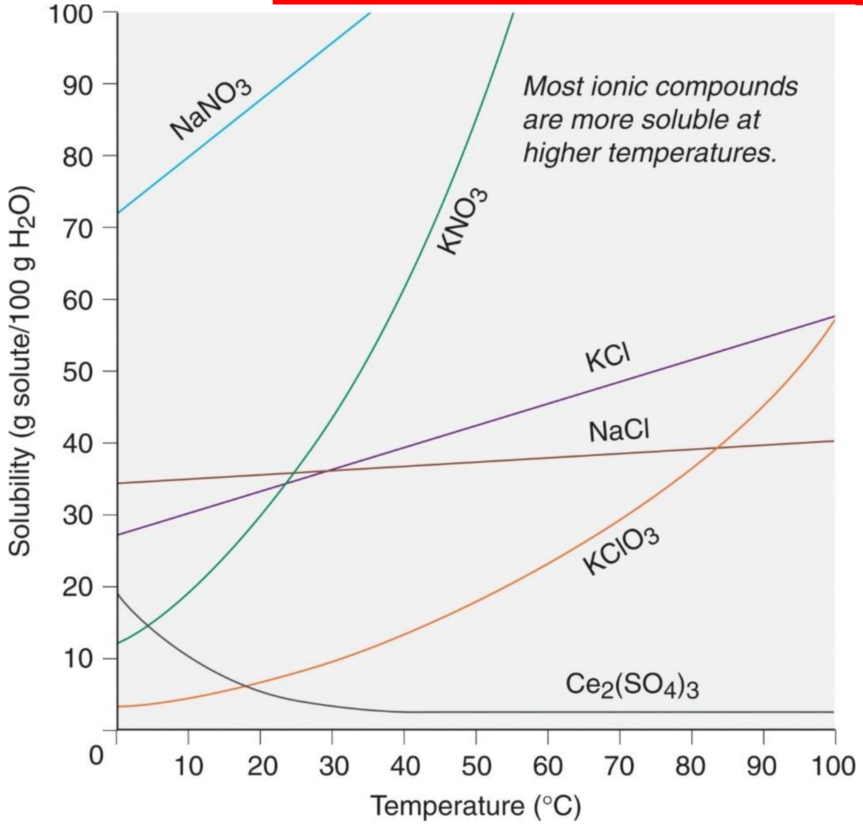
Some Water in Lots of Bromine

Figure 11.15 Bromine (the deep orange liquid on the left) and water (the clear liquid in the middle) are partially miscible. The top layer in the mixture on the right is a saturated solution of bromine in water; the bottom layer is a saturated solution of water in bromine. (credit: Paul Flowers)

Solubility vs Temperature

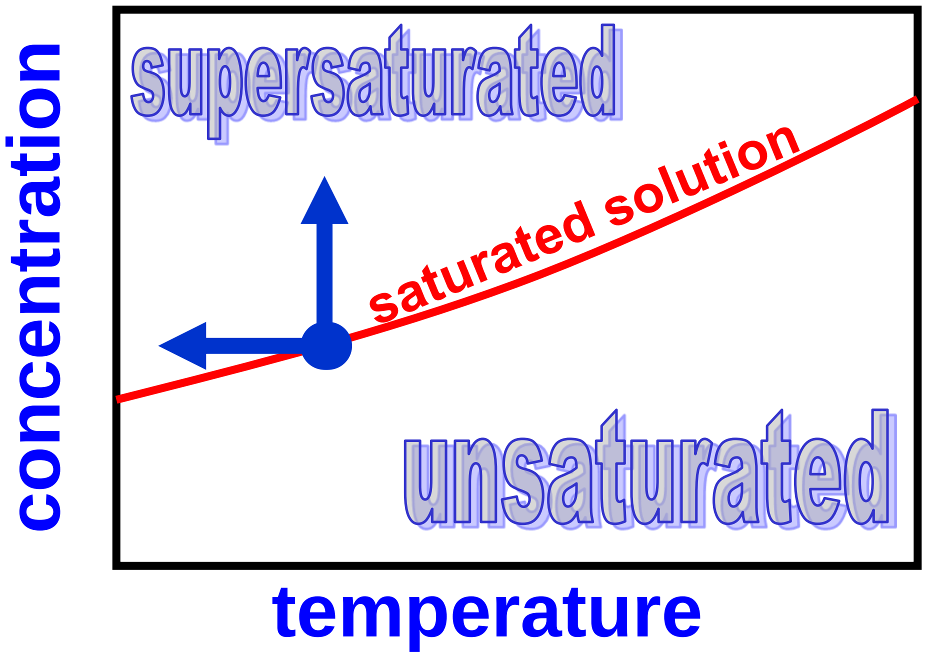
- Solubility increases with increasing temperature for most ionic solids (note exceptions exist)
- In these cases, dissolution is endothermic.

$$\boxed{\textbf{solute + heat} \rightarrow \textbf{solution}}$$

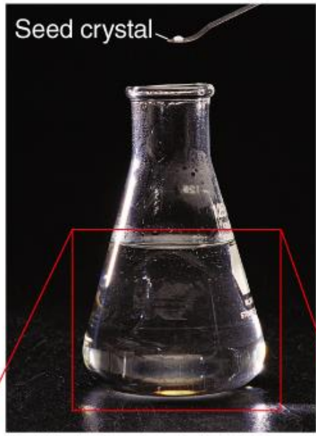


Solubility, cont'd

- A supersaturated solution is obtained by cooling or concentrating a saturated solution.
- Unstable state – spontaneous recrystallization
- Recrystallization is nucleated
  - Must add a few crystals of solute to initiate.



Sodium acetate crystallizing from a supersaturated solution



A

B

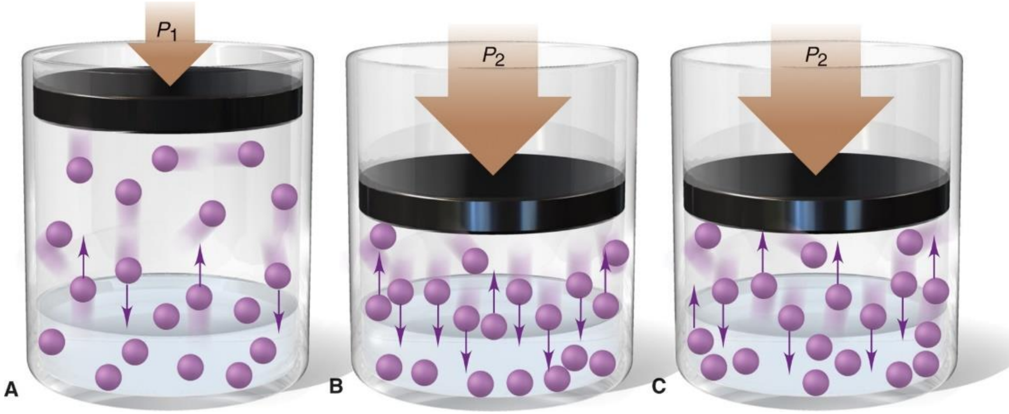
C

A *supersaturated* solution contains *more* than the equilibrium concentration of solute. It is *unstable* and any disturbance will cause excess solute to crystallize immediately.

Figure 11.17 This hand warmer produces heat when the sodium acetate in a supersaturated solution precipitates. Precipitation of the solute is initiated by a mechanical shockwave generated when the flexible metal disk within the solution is "clicked." (credit: modification of work by "Velela"/Wikimedia Commons)

Solubilities of Gases

- Solubility of gases in water generally decreases with increasing temperature.
  - many organisms prefer cooler water.
  - one predicted effect of global warming is a decrease in dissolved oxygen in oceans.



Gas dissolution MUST be exothermic to over come entropic penalty

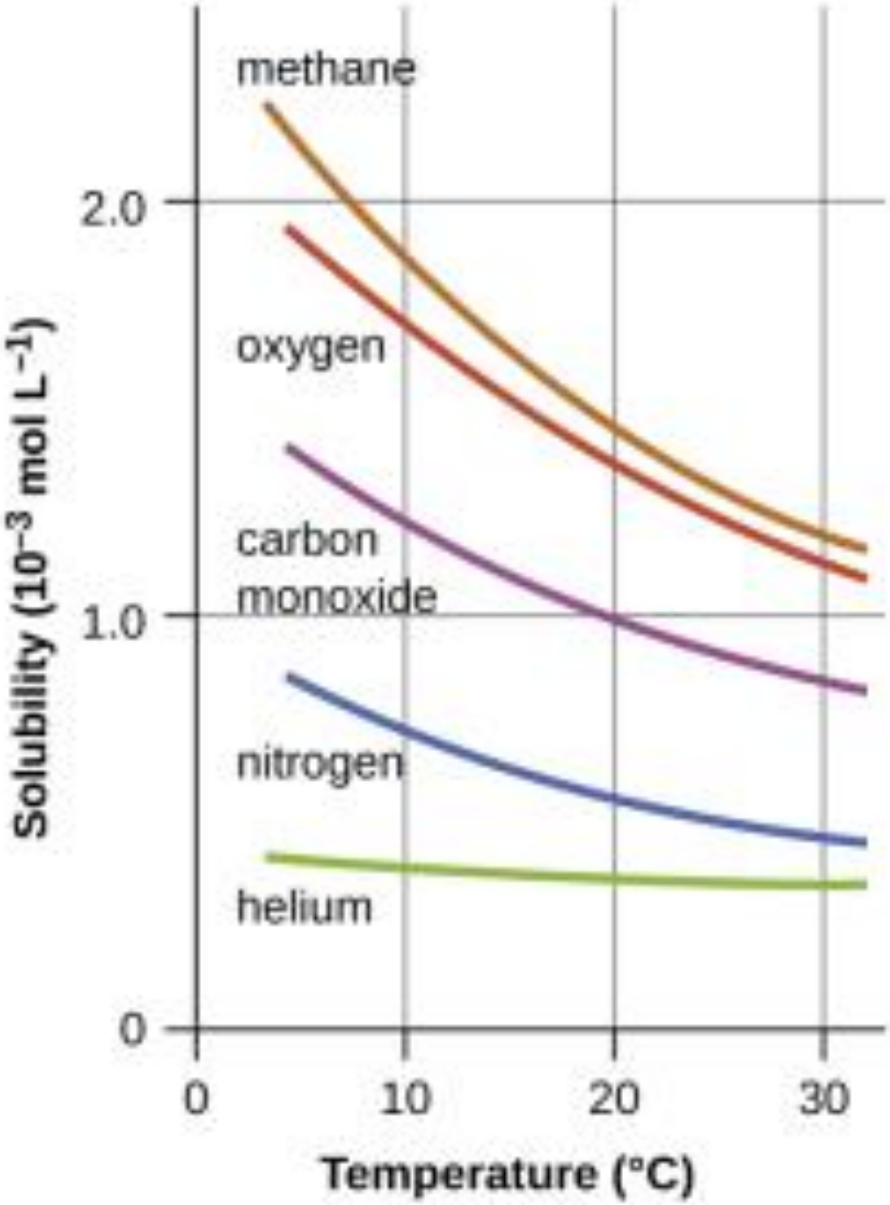


Figure 11.8 The solubilities of these gases in water decrease as the temperature increases. All solubilities were measured with a constant pressure of 101.3 kPa (1 atm) of gas above the solutions.

Solubilities of Gases

William Henry (1774-1836)

- Solubility of a gas increases with increased pressure of the gas above the solvent (1803).
- Established relationship important in early physical chemistry, Henry's Law

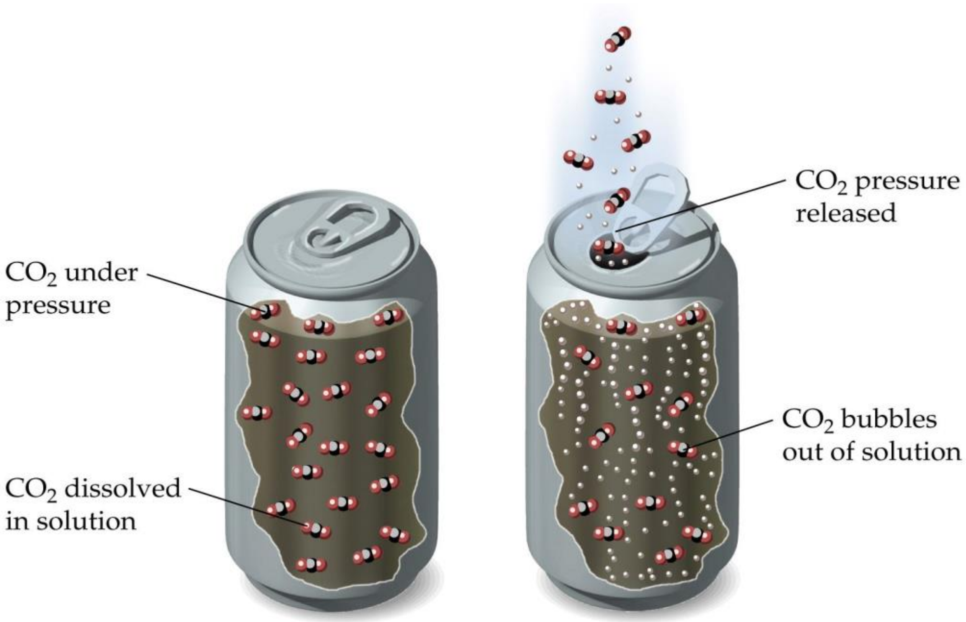
Henry's Law

$$C_{\mathrm{g}} = kP_{\mathrm{g}}$$

Solubility of gas in solution

Henry's Law Constant - Depends on temperature, identity of the gas and the solvent.

Partial pressure of the gas above the solvent.



- Solubility of CO_2 in water is high under high pressure.
- Opening a bottle relieves the pressure, solubility decreases.
- CO_2 evaporates from solution as fizz.